Asymptotic expansion of solutions to Markov renewal equations

Matthias Meiners

May 19, 2026

This talk is based on...

- A. Iksanov, K. Kolesko, M. M.
  Asymptotic fluctuations of supercritical general branching processes.
  Ann. Probab. **52** (2024), no. 4, 1538–1606.
- K. Kolesko, M. M., I. Tomic
  Asymptotic expansions of solutions to Markov renewal equations and their application to general branching processes.
  Submitted to *Journal of Mathematical Analysis and Applications.*

The general branching process

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- Ancestor $\varnothing$ is born at time $t = 0$ with type $\tau(\varnothing) \in \{1, \ldots, p\}$.
- If the ancestor's type is i, then its type-j offspring is born at the times of a point process ξ^{ij}. Write $\xi = (\xi^{ij})_{i,j}$ and

$$\xi = \sum_{j=1}^{p} \xi^{\tau(\varnothing),j} = \sum_{k=1}^{N} \delta_{X_k}.$$

The general branching process

- Associated with each potential individual u there is an independent copy ξ_u of ξ. If Individual u is born at time $S(u) \geq 0$ with type $\tau(u)$, then we set

$$\xi_u = \sum_{j=1}^{p} \xi_u^{\tau(u),j} = \sum_{k=1}^{N_u} \delta_{X_{uk}}.$$

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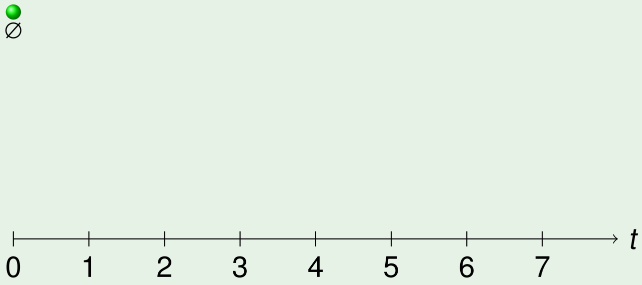
- The kth child of individual u is born at time $S(u) + X_{uk}$.
- The general branching process at time t is

$$\mathcal{Z}_t = \sum_u \mathbf{1}_{\{S(u) \leq t\}}.$$

The general branching process



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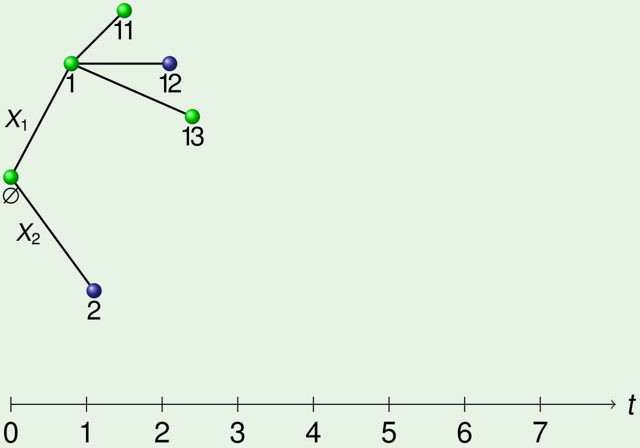

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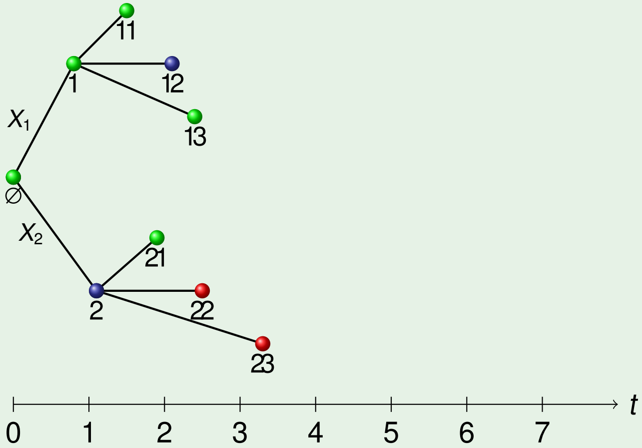

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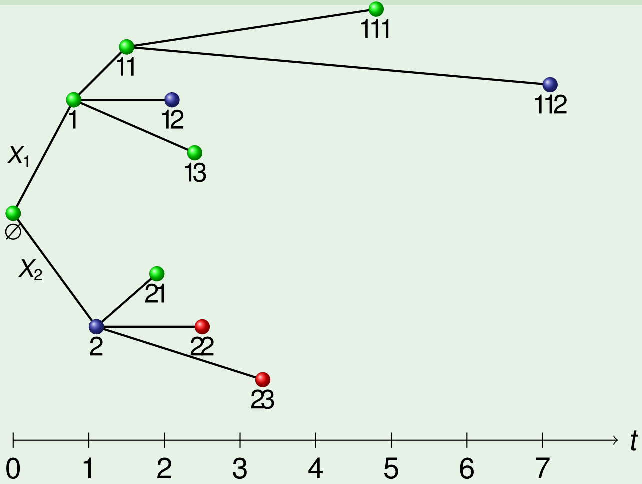

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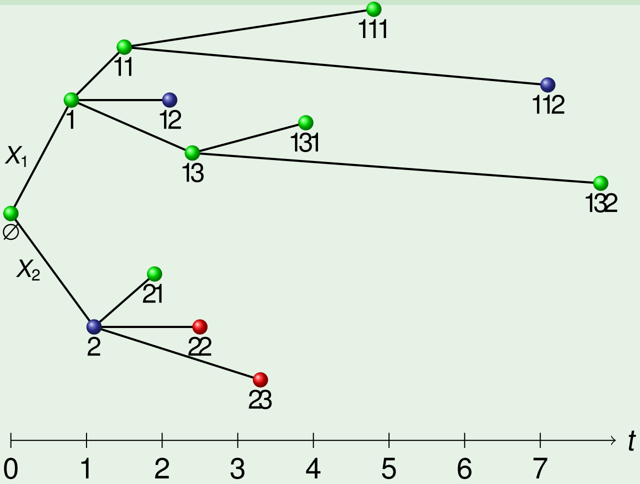

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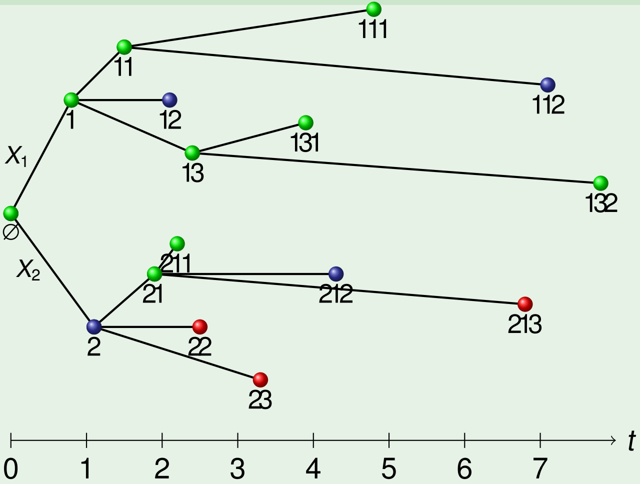

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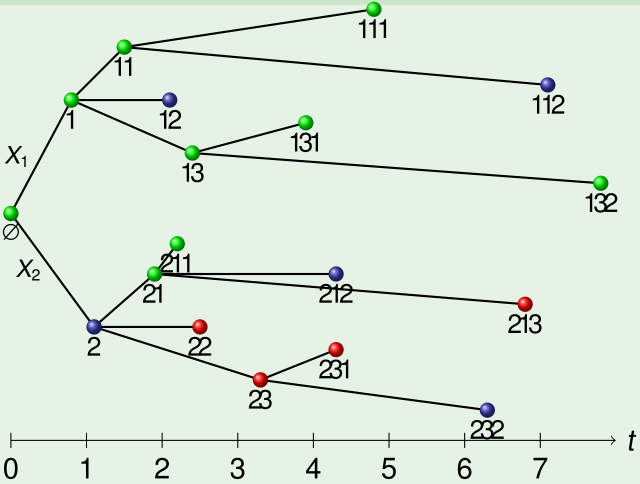

GBP counted with random characteristic

Suppose that for $\forall u \; \exists \; p$-dimensional stochastic process

$$\varphi_u = (\varphi_u(t))_{t \in \mathbb{R}}$$

with càdlàg paths such that (ξ_u, φ_u), u are i. i. d.

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$$\mathcal{Z}_t^{\varphi} = \sum_u \mathrm{e}_{\tau(u)}^{\mathsf{T}} \varphi_u(t - S(u))$$

where $\mathrm{e}_1, \ldots, \mathrm{e}_p$ is the canonical basis of $\mathbb{R}^p$ is called general branching process counted with (random) characteristic φ.

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- $\varphi_u = \mathrm{e_j} \mathbf{1}_{[0,\infty)}$. $\Rightarrow \mathcal{Z}_t^{\varphi} = \#\{u : \tau(u) = j, \, S(u) \leq t\}$.

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- The process was designed to incorporate more features and unify earlier models of branching processes (which I am not going to list here).
- General branching processes are by now classical, but still have various applications.

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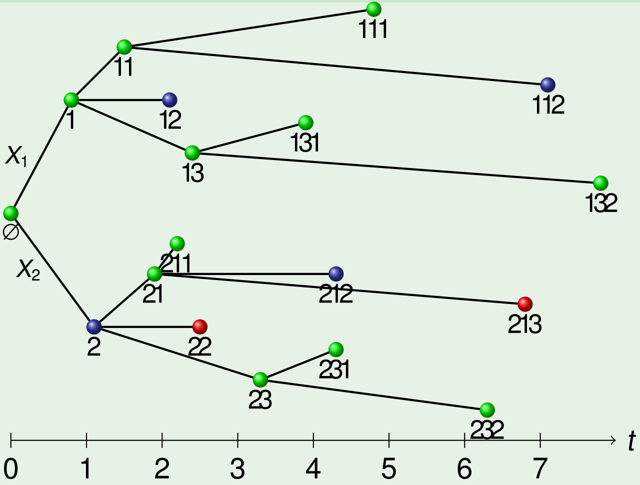
  throughout the talk, we assume that μ is not concentrated on any lattice
- $\mathcal{L}\mu$: matrix of Laplace transforms of μ;
  $\mathcal{L}\mu(\theta) = \left(\int e^{-\theta x} \, \mu^{ij}(\mathrm{d}x) \right)_{i,j} = \left(\mathbb{E}^i \left[\sum_{k:\tau(k)=j} e^{-\theta X_k} \right] \right)_{i,j}$.

The coming generation at time $t = 3$



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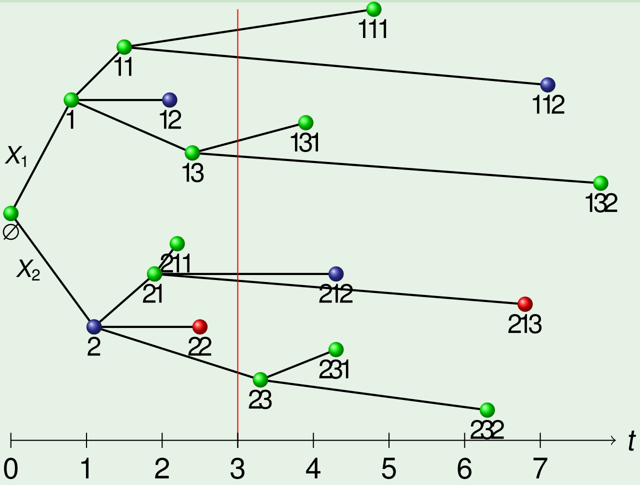

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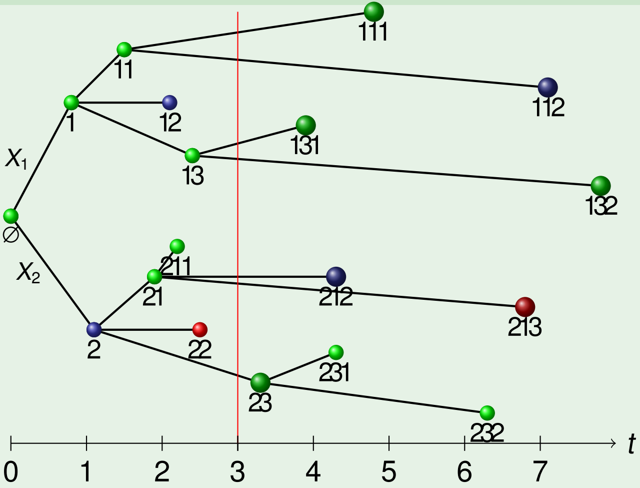

Nerman's martingale

The coming generation at time t:

$$C_t = \{uk : S(u) \leq t < S(uk)\}.$$

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If $\alpha > 0$ is such that $\mathcal{L}\boldsymbol{\mu}(\alpha)$ has Perron–Frobenius eigenvalue 1 and if $\boldsymbol{v} = (\boldsymbol{v}_1, \ldots, \boldsymbol{v}_p)^{\mathsf{T}} \in \mathbb{R}^p$ is a right eigenvector of $\mathcal{L}\boldsymbol{\mu}(\alpha)$ with $\|\boldsymbol{v}\|_1 = 1$, then

$$W_t = \textstyle\sum_{u \in C_t} \boldsymbol{v}_{\tau(u)} e^{-\alpha S(u)}, \quad t \geq 0$$

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is a nonnegative martingale with limit $W \geq 0$. $W_t \to W$ in L^1 if $\mathcal{L}\mu'(\alpha) \in (-\infty, 0]^{p \times p}$, one entry strictly negative, and, for $i = 1, \ldots, p$,

$$\mathbb{E}^i\Big[\Big(\textstyle\sum_{k=1}^N e^{-\alpha X_k}\Big) \log_+\Big(\textstyle\sum_{k=1}^N e^{-\alpha X_k}\Big)\Big] < \infty. \qquad (Z \log Z)$$

Law of large numbers (Nerman 1979)

In the non-lattice case, under the above Malthusian assumptions and if some mild assumptions on φ hold,

$$e^{-\alpha t} \mathcal{Z}_t^{\varphi} \to \frac{W}{\beta} \int_{\mathbb{R}} e^{-\alpha x} \boldsymbol{u}^{\mathsf{T}} \mathbb{E}[\varphi](x) \, \mathrm{d}x \quad \text{in } L^1(\mathbb{P}^i)$$

for some constant $\beta > 0$ that depends only on $\mathcal{L}\boldsymbol{\mu}$ and a nonnegative left eigenvector $\boldsymbol{u} = (\boldsymbol{u}_1, \ldots, \boldsymbol{u}_p)^{\mathsf{T}}$ to the eigenvalue 1 of $\mathcal{L}\boldsymbol{\mu}(\alpha)$ such that $\boldsymbol{u}^{\mathsf{T}} \boldsymbol{v} = 1$.

Fluctuations of the GBP (Iksanov & Kolesko & M. 2024)

Suppose that $p = 1$, that Malthusian assumptions hold,

- $\mathbb{E}\Big[\Big(\sum_{k=1}^{N} e^{-\vartheta X_k}\Big)^2\Big] < \infty$ for some $\vartheta < \frac{\alpha}{2}$
- and that some assumptions on φ hold.

Then there are constants a_λ and $\sigma \geq 0$ such that, with

$$H(t) := \sum_{\frac{\alpha}{2} < \mathrm{Re}(\lambda) \leq \alpha : \mathcal{L}\mu(\lambda) = 1} a_\lambda e^{\lambda t} W(\lambda),$$

it holds that, as $t \to \infty$,

$$e^{-\frac{\alpha}{2} t}\Big(\mathcal{Z}_t^{\varphi} - H(t)\Big) \xrightarrow{\mathrm{d}} \sigma \sqrt{\frac{W}{\beta}} \mathcal{N} \qquad (1)$$

if there are no roots λ of $\mathcal{L}\mu(\lambda) = 1$ with $\mathrm{Re}(\lambda) = \frac{\alpha}{2}$ and all roots are simple.

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- This has been undertaken with Konrad Kolesko, Alicja Kołodziejska, M. M. and Ivana Tomic.
- The first step is to understand the mean $\mathbb{E}^i[\mathcal{Z}_t^{\varphi}]$ as $t \to \infty$.

The Markov renewal equation

By decomposing along the first generation, we infer for $F(t) = (F_1(t), \ldots, F_p(t))^\mathsf{T}$ where $F_i(t) = \mathbb{E}^i[\mathcal{Z}_t^\varphi]$,

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$$= \mathbb{E}^i\Big[\varphi^i(t) + \sum_{k=1}^{N} \sum_{u \in \mathcal{I}} \mathrm{e}_{\tau(ku)}^{\mathsf{T}} \varphi_{ku}(t - X_k - S_k(u))\Big]$$

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&= \mathbb{E}^i\Big[\varphi^i(t) + \sum_{k=1}^N \sum_{u \in \mathcal{I}} \mathrm{e}_{\tau(ku)}^\mathsf{T} \varphi_{ku}(t - X_k - S_k(u))\Big] \\
&= \mathbb{E}^i[\varphi^i(t)] + \sum_{j=1}^p \int F_j(t-x)\, \mu^{ij}(\mathrm{d}x) = f_i(t) + (\mu * F(t))_i
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$$= \mathbb{E}^i[\varphi^i(t)] + \sum_{j=1}^p \int F_j(t-x)\, \mu^{ij}(\mathrm{d}x) = f_i(t) + (\boldsymbol{\mu} * F(t))_i$$

with $f_i(t) = \mathbb{E}[\varphi^i(t)]$.

The Markov renewal equation: Matrix form

In matrix notation, we arrive at

$$F(t) = f(t) + \boldsymbol{\mu} * F(t) \tag{2}$$

for (column) vector-valued functions $f, F : \mathbb{R} \to \mathbb{R}^p$ and the $p \times p$ matrix

$$\boldsymbol{\mu} = \begin{pmatrix} \mu^{11} & \mu^{12} & \cdots & \mu^{1p} \\ \mu^{21} & \mu^{22} & \cdots & \mu^{2p} \\ \vdots & \ddots & \ddots & \vdots \\ \mu^{p1} & \mu^{p2} & \cdots & \mu^{pp} \end{pmatrix}$$

of locally finite measures.

Markov renewal theorem gives the leading order of solutions:

K. S. Crump.
On systems of renewal equations.
J. Math. Anal. Appl., 30:425–434, 1970.

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On systems of renewal equations: The reducible case.
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There are further extensions to a general type space.
However, we are interested in a full expansion as in

M. S. Sgibnev.
Semimultiplicative estimates for the solution of the multidimensional renewal equation.
Izv. Ross. Akad. Nauk Ser. Mat., 66(3):159–174, 2002.

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and notice that in the CMJ context

$$U^{ij}(t) = \mathbb{E}^i[\mathcal{Z}_t^j]$$

with $\mathcal{Z}_t^j = \sum_{u \in \mathcal{I}} \mathrm{e}_{\tau(u)}^{\mathsf{T}} \mathrm{e}_j \mathbf{1}_{[0,\infty)}(t - S(u))$ (the number of individuals of type j born up to time t).

A1 There exists a $\vartheta \in \mathbb{R}$ such that, for some $\theta < \vartheta$ and all i, j,

$$\mathcal{L}\mu^{ij}(\theta) = \int e^{-\theta x} \, \mu^{ij}(\mathrm{d}x) < \infty.$$

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(A2) The spectral radius $\rho_{\boldsymbol{\mu}(0)}$ of $\boldsymbol{\mu}(0) = (\mu^{ij}(0))_{i,j}$ satisfies $\rho_{\boldsymbol{\mu}(0)} < 1$.

Basic facts about the renewal equation

Let μ be a $p \times p$ matrix of locally finite measures on $[0, \infty)$ satisfying (A2) with associated Markov renewal measure $\boldsymbol{U}$. Then the following assertions hold.

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- a The Markov renewal function $\boldsymbol{U}(t)$ is finite at every $t \geq 0$.
- b If (A1) and (A2) hold, then $\rho_{\mathcal{L}\mu(\theta)} < 1$ for all sufficiently large $\theta \in \mathbb{R}$ and, for any such θ, $U^{ij}(t) = o(e^{\theta t})$ for all i, j.

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(c) Let $f = (f_1, \ldots, f_p)^{\mathsf{T}}$ be locally bounded and supported on $[0, \infty)^p$. Then there is a unique locally finite solution $F : \mathbb{R} \to \mathbb{R}^p$ to $F = f + \mu * F$, namely,

$$F(t) = \boldsymbol{U} * f(t), \quad t \in \mathbb{R}.$$

Asymptotic expansion of the renewal function

Recall that

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Take Laplace transforms to infer

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so that

$$\mathcal{L}\boldsymbol{U}(z) = \frac{1}{z}(\mathrm{I}_p - \mathcal{L}\boldsymbol{\mu}(z))^{-1}, \quad \det(\mathrm{I}_p - \mathcal{L}\mu(z)) \neq 0.$$

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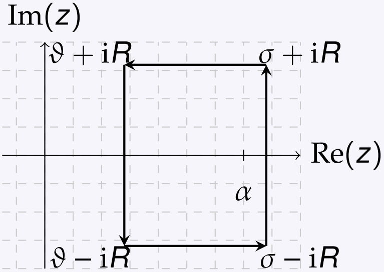
Suppose that $\limsup_{\eta \to \infty} \|\mathcal{L}\boldsymbol{\mu}(\vartheta + \mathrm{i}\eta)\| < 1$. Then $|\Lambda_\vartheta| < \infty$:



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$$
\begin{aligned}
\boldsymbol{U}(t) &= \frac{1}{2\pi \mathrm{i}} \int_{\sigma-\mathrm{i}\infty}^{\sigma+\mathrm{i}\infty} e^{tz} \mathcal{L}\boldsymbol{U}(z)\,\mathrm{d}z \\
&= \sum_{\lambda \in \Lambda_\vartheta} \mathrm{Res}_{z=\lambda}(e^{tz}\mathcal{L}\boldsymbol{U}(z)) + \underbrace{\frac{1}{2\pi \mathrm{i}} \int_{\vartheta-\mathrm{i}\infty}^{\vartheta+\mathrm{i}\infty} e^{tz} \mathcal{L}\boldsymbol{U}(z)\,\mathrm{d}z}_{=O(te^{\vartheta t})}.
\end{aligned}
$$

Asymptotic expansion of the renewal function

$$
\begin{aligned}
\boldsymbol{U}(t) &= \frac{1}{2\pi\mathrm{i}} \int_{\sigma-\mathrm{i}\infty}^{\sigma+\mathrm{i}\infty} e^{tz} \mathcal{L}\boldsymbol{U}(z)\,\mathrm{d}z \\
&= \sum_{\lambda \in \Lambda_\vartheta} \mathrm{Res}_{z=\lambda}(e^{tz}\mathcal{L}\boldsymbol{U}(z)) + \underbrace{\frac{1}{2\pi\mathrm{i}} \int_{\vartheta-\mathrm{i}\infty}^{\vartheta+\mathrm{i}\infty} e^{tz} \mathcal{L}\boldsymbol{U}(z)\,\mathrm{d}z}_{=O(te^{\vartheta t})}.
\end{aligned}
$$

After some computations, we obtain

$$
\boldsymbol{U}(t) = \sum_{\lambda \in \Lambda_\vartheta} e^{\lambda t} \sum_{k=0}^{\mathrm{k}(\lambda)-1} t^k C_{\lambda,k} + t^{\mathrm{k}(0)} C_{0,\mathrm{k}(0)} + O(te^{\vartheta t}) \quad \text{as } t \to \infty.
$$

Asymptotic expansion of solutions to Markov renewal equations

Let $f : [0, \infty) \to \mathbb{R}^p$ be càdlàg with the property

$$\int_0^\infty e^{-\theta x} \, \mathrm{V}f(x) \, \mathrm{d}x < \infty$$

for some $\theta > \vartheta$ with $\Lambda_\vartheta = \Lambda_\theta$. Then, for $F = f + \mu * F$,

$$F(t) = \sum_{\lambda \in \Lambda_\theta} e^{\lambda t} \sum_{k=0}^{\mathrm{k}(\lambda)-1} t^k b_{\lambda,k,f} + O(e^{(\theta+\varepsilon)t})$$

as $t \to \infty$, where $\varepsilon > 0$ can be chosen arbitrarily small.

Thank you for your attention